

CYPRESS RIVER, MB, Canada

WMO: 712390

Lat: 49.5622N

Lon: 99.0739W

Elev: 374

StdP: 96.91

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-28.5	-34.6	0.1	-31.0	-32.0	0.2	-28.3	12.6	-11.5	11.3	-8.4	2.9	220	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	30.4	20.1	28.5	19.5	26.9	18.7	22.3	27.7	21.1	26.5	19.9	25.0	4.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	15.9	25.6	19.2	14.6	24.5	18.0	13.6	22.8	67.7	27.6	63.2	26.6	58.9	24.8	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.4	8.5	-34.5	34.7	2.9	1.7	-36.6	35.9	-38.3	37.0	-39.9	37.9	-42.0	39.2		
(5)			-34.6	25.0	2.9	1.2	-36.7	25.8	-38.3	26.5	-39.9	27.2	-42.0	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-14.1	-14.0	-5.3	3.4	11.5	16.9	19.4	18.5	13.3	5.4	-3.9	-12.4		
	DBStd	13.74	8.42	7.53	8.11	6.10	5.09	3.19	2.87	3.58	4.76	5.24	7.05	7.82		
	HDD10.0	3443	748	673	474	207	43	1	0	0	21	163	417	696		
	HDD18.3	5621	1006	906	731	447	219	63	21	44	161	401	666	954		
	CDD10.0	1004	0	0	2	10	89	207	292	262	120	21	1	0		
	CDD18.3	140	0	0	0	0	7	19	55	48	11	1	0	0		
	CDH23.3	1583	0	0	0	1	141	208	532	550	142	9	0	0		
	CDH26.7	419	0	0	0	0	36	36	122	184	40	2	0	0		
(14) Wind	WSAvg	4.0	4.2	4.0	4.1	4.6	4.5	3.9	3.3	3.3	3.9	4.3	4.3	3.9		
(15) Precipitation	PrecAvg	510	19	17	22	30	75	89	73	59	41	38	21	25		
	PrecMax	679	44	60	53	60	166	189	156	106	99	120	78	44		
	PrecMin	395	1	1	0	5	23	38	25	14	5	2	2	5		
	PrecStd	78	10	14	16	14	30	33	27	26	20	25	17	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	4.2	18.0	22.8	30.5	29.6	31.5	33.7	30.9	24.1	15.2	4.7		
		MCWB	1.4	1.7	11.0	11.3	16.0	19.7	22.7	20.8	19.2	15.0	9.9	1.7		
	2%	DB	2.3	2.1	12.7	19.0	26.8	27.4	29.4	30.9	27.2	20.2	10.7	1.6		
		MCWB	0.5	0.6	7.8	9.8	14.5	18.0	21.4	19.9	17.2	12.1	6.5	0.0		
	5%	DB	0.5	0.2	8.2	16.5	24.4	25.7	28.0	28.5	24.3	17.0	7.6	-0.2		
		MCWB	-0.6	-1.0	5.3	8.5	13.3	17.0	20.5	19.5	16.0	10.9	4.5	-1.3		
	10%	DB	-1.7	-2.4	4.6	13.7	21.4	24.1	26.3	26.4	21.5	13.8	4.9	-2.1		
		MCWB	-2.7	-3.4	2.5	7.5	12.2	16.5	19.5	18.9	14.9	9.4	2.9	-2.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	2.2	13.1	12.2	18.5	22.2	24.6	23.2	21.5	15.7	10.6	2.0		
		MCDB	3.4	4.2	15.4	20.0	26.8	26.2	28.9	29.5	27.6	22.3	15.6	4.4		
	2%	WB	0.7	0.6	7.6	10.4	16.7	19.9	22.7	21.8	19.0	13.5	6.8	0.1		
		MCDB	2.1	1.8	11.8	17.3	21.8	24.4	27.7	27.8	23.8	18.6	9.8	1.5		
	5%	WB	-0.7	-1.0	4.9	8.9	14.8	18.6	21.5	20.8	17.2	11.5	4.7	-1.2		
		MCDB	0.4	0.2	8.3	15.6	21.1	23.5	26.5	26.6	22.2	16.4	7.2	-0.2		
	10%	WB	-2.7	-3.4	2.5	7.6	13.4	17.5	20.3	19.5	15.8	9.5	3.0	-2.8		
		MCDB	-1.8	-2.3	4.6	13.1	19.5	22.3	25.1	25.0	20.5	13.6	4.8	-2.1		
(35) Mean Daily Temperature Range	MDBR		9.4	10.1	10.2	11.7	12.5	11.4	11.9	13.4	12.1	10.1	8.7	8.3		
	5% DB	MCDBR	9.9	10.3	12.9	16.8	17.3	14.0	14.0	16.5	16.5	15.2	12.3	9.5		
		MCWBR	8.8	8.6	8.7	9.0	7.8	7.1	7.5	7.8	8.0	8.7	8.4	8.2		
	5% WB	MCDBR	10.2	10.1	12.7	15.0	13.1	11.7	12.4	14.0	13.0	13.6	11.4	9.1		
		MCWBR	9.1	8.6	8.7	8.6	6.7	7.0	7.5	7.3	7.5	8.0	8.0	7.9		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.271	0.291	0.351	0.366	0.373	0.382	0.380	0.345	0.295	0.266	0.242		
	taud		2.282	2.271	2.384	2.334	2.350	2.369	2.351	2.349	2.432	2.545	2.480	2.391		
	Ebn at Noon		794	879	926	895	892	885	871	854	853	854	802	784		
	Edh at Noon		69	90	97	115	118	117	117	112	93	69	57	54		
(44) All-Sky Solar Radiation	RadAvg		1.47	2.65	3.92	5.00	5.54	5.96	6.33	5.20	3.79	2.31	1.30	1.05		
	RadStd		0.12	0.20	0.31	0.34	0.44	0.40	0.26	0.35	0.28	0.30	0.18	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	-0.68	N/A	N/A	N/A	N/A

Nomenclature: See separate page